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### Capsule adapted to receive a fluid, and assembly comprising a container and said capsule

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#### Abstract

A capsule to seal the opening in a container, such as the neck of a bottle, wherein the container is adapted to contain a first fluid, the capsule being adapted to contain a second fluid that is different from the first fluid, the capsule including a striker movable from a first initial position to a second opening position and which, in the second position, allows the second fluid to be released into the container, thus mixing the first fluid and the second fluid, the striker being in the form of a cylinder adapted to allow a fluid discharge device, such as a pipette, to pass through the capsule and to enter the container.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims priority to PCT/EPFR2020/050576 filed Mar. 17, 2020 which claims priority to French Patent Application No. 1903024 filed Mar. 22, 2019, the entire disclosures of which are incorporated by reference herein.

## TECHNICAL FIELD

(1) The invention relates to a capsule adapted to seal the opening in a container, such as the neck of a bottle, wherein said container is adapted to contain a first fluid, the capsule being, for its part,

adapted to contain a second fluid which is different from said first fluid.

(2) The capsule according to the present invention is provided with a space intended for receiving at least one second fluid, which is different from the first fluid. The invention also relates to the assembly made up of a container, which is adapted to package a first fluid, and a capsule according to the invention.

## BACKGROUND

(3) Capsules which are adapted to provide a space that is intended to contain a second fluid have already been disclosed in the prior art, said capsules being adapted to seal a container envisaged for containing a first fluid. At a moment determined by the user, the contents of the capsule, namely the second fluid contained in said capsule, can be released in order to mix with the first fluid that is present in the container. An example of a capsule of this type is described in U.S. Pat. No.

8,413,802. This type of capsule can prove to be useful for mixing a first and a second fluid, just before the container is first opened, in order to ensure that said first fluid and said second fluid are freshly mixed before said mixture is expelled from the container. Another option for this type of capsule consists of offering the user the choice of mixing or not mixing the contents of the capsule, i.e. the second fluid, with the first fluid contained in the container.

(4) The capsules disclosed in the state of the art exhibit a special characteristic according to which, if said capsule is removed from the container in order to be opened, the space intended for containing the second fluid is removed from said container at the same time. A first disadvantage results from the fact that if the mechanism which lets the second fluid be mixed with the first fluid, which is present in the container, has been used before, the space intended for accommodating the second fluid has already been opened. By taking the capsule out of the container, the rest of said second fluid can possibly remain, in a pure state, accessible for the user. If the second fluid is particularly acidic or corrosive, contact with the second fluid, in a pure state, can pose a risk for the user.

(5) Moreover, by removing the space intended for receiving the second fluid, when the capsule according to the prior art is removed from the container, the side of the space, which is adapted for receiving said second fluid, in a position in which the capsule is orientated towards the interior of the container, is exposed and can be in contact with any type of contamination. Consequently, after a potentially contaminated capsule is repositioned, the purity of the mixture of the first and the second fluid inside the container can be seriously compromised.

(6) Another disadvantage of the capsule disclosed in the prior art is the fact that it does not provide any support to a fluid extraction device such as a pipette. In the event that the first fluid and the second fluid to be packaged in the assembly made up of the device and the capsule are pharmaceutical products or cosmetics, it can be advantageous to use a pipette in order to extract the first fluid and/or the mixture of the first and the second fluid from the interior of the container to the exterior.

(7) With regard to the observations above, it appears that there is a need to improve the existing containers which are intended to contain a first fluid and which can cooperate with a capsule adapted to pour a second fluid and mix it with said first fluid.

## SUMMARY

(8) The subject of the invention is a capsule adapted to seal the opening in a container, such as the neck of a bottle, wherein said container is adapted to contain a first fluid, the capsule being adapted to contain a second fluid that is different from said first fluid, said capsule comprising a striker which is movable from a first initial position to a second opening position and which, in said second position, allows said second fluid to be released into the container, thus mixing the first fluid and the second fluid, said striker being in the form of a cylinder adapted to allow a fluid discharge device, such as a pipette, to pass through the capsule and to enter the container.

(9) According to one embodiment of the invention, said capsule comprises fixing structure(s), such as a rib, for fixing the capsule in the opening in the container.

- (10) According to one embodiment of the invention, said capsule, in the use position of said capsule, on the side exposed towards the interior of the container, is delimited by a first protective cover, wherein the striker is adapted, in its second opening position, to open said protective cover.
- (11) According to one embodiment of the invention, the striker comprises, in the use position of said capsule, at the end directed towards the exterior of the container, a transverse wall adapted to cooperate with a push-piece which thus allows said striker to be moved forward from its initial position to its opening position.
- (12) According to one embodiment of the invention, said transverse wall comprises at least part of the outer wall of said capsule, in its use position.
- (13) According to one embodiment of the invention, the wall of said capsule exposed towards the exterior of the container, in the use position of the capsule, is made of a flexible material on which the striker is fixed.
- (14) According to one embodiment of the invention, the striker in the use position of the capsule, on the side exposed towards the exterior of the container, is sealed with the aid of a second protective cover.
- (15) According to one embodiment of the invention, said second protective cover is adapted to be struck with the aid of a fluid discharge device such as a pipette.
- (16) According to one embodiment of the invention, said second protective cover is adapted to be taken off.
- (17) According to one embodiment of the invention, the first protective cover comprises a material which is impermeable to oxygen, such as aluminum.
- (18) The present invention also relates to an assembly comprising a container, adapted to contain a first fluid, with an opening which allows said container to be filled and emptied, a lid adapted to seal said opening and a capsule described above, wherein the container is provided with fixing structure such as a thread which allows said lid to be fixed onto the container, wherein the interior of said lid is provided with at least one first push-piece adapted to push the capsule towards the interior of the opening in the container, thus fixing said capsule in said opening.
- (19) According to one embodiment of the invention, the interior of the lid is provided with at least one second push-piece adapted to cooperate with the transverse wall of the striker, thus allowing said striker to be moved forward from its first initial position to its second opening position.
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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The aim, subject and characteristics of the invention will become more clearly apparent on reading the following description with reference to the figures, in which:
- (2) FIG. 1 shows a container adapted to cooperate with the capsule according to the invention, said container being provided with a protection member;
- (3) FIG. 2 depicts a pipette lid adapted to be fixed onto the container according to FIG. 1;
- (4) FIG. 3 illustrates the fixing, by screwing, of the pipette lid, according to FIG. 2, onto the container according to FIG. 1;
- (5) FIG. 4 shows the pipette lid fixed on the container after the screwing step, as shown in FIG. 2;
- (6) FIG. 5 depicts a diagrammatic view of the sampling of a specific quantity of fluid, contained within the container, with the aid of the pipette, said pipette forming an integral part of the pipette lid;
- (7) FIG. 6 illustrates the sample return step performed with the aid of the pipette, said pipette forming an integral part of the pipette lid;
- (8) FIG. 7 shows a sectional view of the fixing of the pipette lid on the container according to FIG. 3;

- (9) FIG. 8 depicts a detailed view of the contact between the upper edge of the container, the body of the pipette lid and the capsule, during the step of fixing said pipette lid onto said container;
- (10) FIG. 9 illustrates a sectional view of the fixing of the pipette lid onto the container, after the capsule is fully inserted inside said container;
- (11) FIG. 10 shows a detailed view of the contact between the upper edge of the container, the body of the pipette lid and the capsule, at the moment when said capsule is inserted inside said container during the fixing of said pipette lid onto the container;
- (12) FIG. 11 depicts a diagrammatic view of the mixing between the first fluid contained in the container and the second fluid released from the capsule;
- (13) FIG. 12 illustrates a sectional view of the sampling of a specific quantity of fluid performed with the aid of the pipette;
- (14) FIG. 13 depicts an embodiment of a capsule according to the invention, in a top view;
- (15) FIG. 14 shows the capsule according to FIG. 1, in a bottom view;
- (16) FIG. 15 depicts a sectional view of an assembly made up of a container and a capsule fixed on the outlet orifice of said container;
- (17) FIG. 16 illustrates the moment when the outer protective cover is removed; and
- (18) FIG. 17 depicts the introduction of a graduated sampling pipette towards the interior of the container.

#### DETAILED DESCRIPTION

- (19) FIG. 1 shows a container 1 adapted to cooperate with the capsule according to the present invention. According to FIG. 1, the container 1 is provided with a protection member 2 which must be taken off before said container 1 is used. Once the protection member 2 has been taken off, a pipette lid 3, as shown in FIG. 2, can be fixed on said container 1. The interior of the pipette lid 3 comprises a capsule 10 according to the present invention. During the first fixing of the pipette lid 3 onto the container 1, the capsule 10 is simultaneously introduced into the interior of said container 1. As described in more detail below, the capsule 10 is adapted to seal the opening in the container 1 and is provided with fixing structure(s) which allow it to be held once it is introduced into the container 1. In other words, the capsule 10 is transferred from the pipette lid 3 to the container 1 during its first fixing onto the container 1.
- (20) The fixing of the pipette lid 3 onto the container 1 is depicted in FIG. 3. Firstly, the pipette lid 3 is positioned on the upper edge 11 of said container 1. A thread 12 present on the exterior of the container 1 can then be used, in combination with a thread present inside the pipette lid 3 (as shown in FIG. 2), to screw said pipette lid 3 onto the container 1. During this screwing step, as explained above, the capsule 10 is fixed inside the container 1 next to the upper edge 11 of said container 1.
- (21) As illustrated in FIGS. 2 and 3, the pipette lid 3 is provided with an integral pipette 20, which pipette 20 is provided with a rod 21 adapted to enter into the container 1. The pipette 20 is also provided with a head 22 which allows the user to take a sample of a specific quantity of product present inside the container 1.
- (22) FIG. 4 illustrates the pipette lid 3 fixed on the container 1. As depicted in FIG. 4, the user has direct access to the head 22 of the pipette 20 which can lower a plunger 24 (see FIG. 7) of the pipette 20, from its position as shown in FIG. 4 in the direction of the position as shown in FIG. 5. When the plunger 24 is pushed in, the user can let go of the head 22 of said pipette 20 so that it again takes up the position as illustrated in FIG. 4. This movement allows the pipette 20 to be filled with a specific quantity of fluid. After the pipette 20 is filled, the user can unscrew the pipette lid 3 and allow the return of the sampling performed by pressure on the head 22 of the pipette 20. FIG. 6 offers a diagrammatic illustration of the return of a quantity of fluid.
- (23) According to the examples illustrated in FIGS. 4, 5 and 6, the plunger 24 of the pipette 20 can be displaced by approximately 5 to 10 millimetres between the initial position depicted in FIG. 4 and the final position depicted in FIG. 5. According to the example illustrated in FIGS. 1 to 12, such a displacement corresponds to a quantity of fluid of approximately 1 to 2 ml sampled from the

container. This indicates that the user has, after a sampling, a specific and relatively limited quantity in the pipette **20** before the return of said limited quantity. An overdosage of fluid is thus ensured with the aid of said pipette **20**. It should be understood that a larger or smaller displacement is possible depending on the user's specific needs and the characteristics of the fluid to be sampled in the container **1**.

(24) The examples depicted in FIGS. **7** to **12** illustrate, in a more detailed manner, the functioning of the device described above.

(25) FIG. **7** depicts a sectional view of the assembly made up of the container **1** and the pipette lid **3**. FIG. **7** illustrates a first screwing of the pipette lid **3** onto the container **1**. Firstly, the capsule **10** situated inside the body **30** of the pipette lid **3** comes into contact with the container **1** next to the edge **11**, which is illustrated in detail in FIG. **8**. The capsule **10** is provided with an assembly of ribs **41**, **42**, **43** which have the purpose of coming into contact with the inner wall of the container **1** next to the edge **11**. The elastic deformation of said ribs **41**, **42**, **43** ensures a force which allows the capsule **10** to be fixed inside the container **1**. The vertical movement of said capsule **10** towards the interior of the container **1** is limited by the stop **44** adapted to come into contact with the edge **11** during the entry of the capsule **10** into the container **1**. The first fixing of the pipette lid **3** onto the container **1** is ensured with the aid of screwing, which is performed with the aid of the threads present on the interior of the body **30** of the pipette lid **3** on the one hand and, on the other hand, on the exterior of the container **1**, below the edge **11**. By screwing the pipette lid **3** onto the container **1**, the capsule **10** located inside the body **30** of the pipette lid **3** is fixed inside the container **10**, as is described above.

(26) The pipette lid **3** is provided with a pipette **20** which forms an integral part of said pipette lid **3**. The pipette **20** is provided with a head **22** adapted to exert pressure which makes it possible to push in a plunger **24** relative to the body **30** in a rod **21**. When the force of pressure ceases, the pipette **20** takes up its initial position again by virtue of the presence of a spring **23**. The pipette **20** comprises a rod **21** which is adapted to extend inside the container **1** in order to come into contact with a fluid **51** present inside said container **1**. The rod **21** is connected to the material of the body **30** and remains immobile relative to said body **30**. The plunger **24** functions as a movable sampling member inside the rod **21**. The plunger **24** can be displaced relative to said rod **21** by virtue of the pressure exerted onto the head **22** of the pipette **20**. The spring **23** is situated between the rod **21** and the plunger **24**, thus allowing, by virtue of the movement of said plunger **24** relative to said rod **21**, the sampling of a specific quantity of fluid with the aid of the pipette **20**.

(27) Before the pipette lid **3** is mounted on the container **1**, as depicted in FIGS. **2** and **7**, the capsule **10** present inside the body **30** of the lid **3** is held in place with the aid of a friction force exerted by the interior of the capsule **10** onto the exterior of the rod **21**. Said friction force is, on the one hand, sufficient to avoid any involuntary displacement of the capsule **10** relative to the rod **22** and, on the other hand, is sufficiently limited to allow the movement of the rod **21** relative to the capsule **10** once said capsule **10** is correctly fixed inside the container **1**.

(28) The capsule **10** has three members which together form a space adapted to contain a second fluid **52** inside said capsule **10**. With reference to FIG. **8**, the member is in the form of a cylindrical body **40**, representing the outer wall of the capsule **10**. As described above, the cylindrical body **40** is provided with ribs **41**, **42**, **43** and a stop **44**. In the position as depicted in FIG. **8**, the cylindrical body **40** is open at its lower end, which lower end is sealed by a protective cover **60**, the protective cover **60** being the second member which forms the space adapted to contain the second fluid **52**.

(29) The third member of the capsule **10** is formed by a cylindrical member **45**, which acts as a striker. The lower part of said striker **45**, i.e. the part directed towards the interior of the container **1** after assembling, is provided with a sharp end **46** adapted to perforate the protective cover **60**, thus releasing the second fluid **52**. The striker **45** is, at its upper end, i.e. the end directed towards the exterior of the container **1** after assembling, connected to a transverse wall **47** forming the upper wall of the capsule **10**. The transverse wall **47** is provided, at its outer end, with a sealing grommet

**48** adapted to come into contact with the inner walls of the cylindrical body **40**. The sealing grommet **48** exerts onto the interior of the cylindrical body **40** a force sufficient to guarantee the seal between said cylindrical body **40** and said sealing grommet **48**. However, the friction force between the cylindrical body **40** and the sealing grommet **48** allows the assembly made up of the striker **45**, the transverse member **47** and the sealing grommet **48** to be displaced relative to the cylindrical body **40**, once a sufficient force is exerted onto the transverse wall **47**.

(30) The lower end of the striker **45** is provided with a member **46** adapted to strike the protective cover **60**. It should be noted that said striker **45** can include a protrusion **91** which allows it to exert, at a specified location, a specific force against the rod **21**. Said protrusion can act as a scraper once the capsule **10** is fixed inside the container **1** and once a user removes the pipette **20** from said container **1**. By virtue of the protrusion **91**, an excess of fluid present on the outer wall of the rod **21** can be removed from said outer wall before the pipette **20** exits the container **1**. The sealing grommet **48** can also be provided with a protrusion **92**, which allows said sealing grommet **48** to exert a force against the inner wall of the cylindrical body **40**.

(31) With reference to FIG. 7, it should be noted that the interior of the body **30** of the lid **3** is provided with an assembly made up of a first push-piece **71** and a second push-piece **72**, this second push-piece **72** being longer than the first push-piece **71**. The first push-pieces **71** are adapted to come into contact with the stop **44** and to move said stop forward against the edge **11**. The second push-pieces **72** are, for their part, adapted to come into contact with the transverse wall **47** in order to displace the assembly made up of the striker **45**, the transverse wall **47** and the sealing grommet **48** relative to the cylindrical body **40**, during the first mounting of the pipette lid **3** onto the container **1**. Thus, the sharp member **46** can perforate the protective cover **60** and release the second fluid **52**, allowing mixing between the first fluid **51**, which is already present inside the container **1**, and the second fluid **52** present in the capsule **10**.

(32) FIG. 9 illustrates the mounting of the container **1** and of the pipette lid **3**, at the moment when the capsule **10** has fully entered inside said container **1**. As depicted in greater detail in FIG. 10, the capsule **10**, which has completely entered inside the container **1**, and the three ribs **41**, **42**, **43** of the cylindrical body **40**, have come into contact with the inner wall of said container **1**, next to the edge **11**. After this first phase, as shown in FIGS. 9 and 10, the user can continue to screw the pipette lid **3** onto the container **1** and, consequently, allow the push-pieces **72** to come into contact with, and exert a force onto, the transverse wall **47**. By pushing the transverse wall **47**, the striker **45** is also displaced, thus allowing the sharp member **46** to perforate the protective cover **60**, as described above.

(33) FIG. 11 depicts the conclusion of the first mounting of the pipette lid **3** onto the container **1**, just before the first fluid **51** and the second fluid **52** mix. The striker **45** has perforated, with the aid of the sharp member **46**, the protective cover **60**, thus releasing the second fluid **52**. It should be understood that this perforation phase is described diagrammatically, the movement of the second fluid **52** in the direction of the first fluid **51** being indicated with the aid of the arrow **80**.

(34) It should be noted that, by virtue of the functioning of the capsule **10** and of the pipette **20**, the fluid **52** cannot be expelled from the container **1** without previously being mixed with the fluid **51**. Consequently, any risk of direct contact between the fluid **52**, in its pure state, and a user's skin is thus avoided.

(35) FIG. 12 depicts a diagrammatic view of the sampling of a specific quantity of fluid **51**, **52** with the aid of the pipette **20**. Pressure is exerted onto the head **22** of the pipette **20** in order to push the plunger **24** in the direction of the base of the container **1**. After the step described in FIG. 12, the user can let go of the head **22** again, thus allowing the plunger **24** to take up its initial position again. The movement of the plunger **24** relative to the rod **21** allows a specific quantity of fluid **51**, **52** to be sampled towards the interior of the rod **21**. After the sampling step, the pipette lid **3** can be removed from the container **1** in order to allow said sample to be returned by pressing on the head **22** of the pipette **20**, as shown in FIG. 6.



(36) FIG. 13 shows a second embodiment of a capsule **101** according to the present invention, which is adapted to be fixed onto the opening in a container, said container itself being adapted to package a first fluid. As explained with reference to FIG. 13, the interior of the capsule **101** is hollow and comprises a space intended for receiving a second fluid.

(37) The capsule **101** is depicted in FIG. 13 in a perspective top view. This indicates that the upper part visible in FIG. 13 remains accessible when said capsule **101** is fixed on the opening in a container.

(38) The capsule **101** comprises an upper wall **110**, provided with an edge **111**, the two being adapted to cooperate with the container on which said capsule **101** is fixed. The edge **111** has a diameter which is essentially similar to the diameter of the opening in the container on which the capsule **101** is fixed. This is depicted more clearly in FIG. 15.

(39) The capsule **101** has a longitudinal wall **120** which is depicted in FIG. 14. Fixing structures **121** are present on said longitudinal wall **120**. According to the example, as depicted in FIG. 13, said fixing structures **121** present on the capsule **101** have the form of circumferential members present on the exterior of the longitudinal wall **120**.

(40) The fixing structures **121** have the special characteristic of deforming, partially elastically, so that the end of the circumferential members exerts a certain force onto the interior of the opening in the container onto which the capsule **101** has been fitted, in order to ensure leaktightness at the end of the fixing structures **121** and inside the opening in said container on which the capsule **101** is fixed.

(41) According to an alternative embodiment (not shown in the figures), the fixing structures **121** can, for example, be in the form of a thread which makes it possible to cooperate with fixing structures present in the opening in a container.

(42) The upper wall **110** is provided with a recess in which the upper part of a striker **140**, which has an essentially cylindrical form, is visible. The form of the striker **140** is more clearly depicted in FIG. 15. The upper end of the striker **140** is sealed with the aid of a protective cover **141** provided with a tongue **142**, which facilitates the removal of said protective cover **141** by the user.

(43) The protective cover **141** comprises a material, or an assembly of materials, specifically adapted to protect the second fluid inside the capsule **101**. This or these material(s) is or are particularly adapted to avoid the second fluid coming into contact with oxygen. Aluminum is a material particularly adapted to the manufacture of the protective cover **141**.

(44) As depicted in FIG. 13, the striker **140** is fixed, close to its upper end, onto a wall **150** made of a flexible material. In the present description, the term “flexible material” indicates that the wall **150** is adapted to be deformed to allow the striker **140** to be displaced from a first to a second position and to allow the storage of a specific quantity of energy, which allows said striker to return to its initial position without altering the integrity of said wall **150**.

(45) FIG. 14 depicts a perspective view of the capsule **101** according to FIG. 13, its lower end being directed upwards. FIG. 14 shows that the lower end of the capsule **101** is sealed with the aid of a protective cover **130**. As depicted in detail in FIG. 15, said protective cover **130** is adapted to seal an interior space inside said capsule **101**, this space being adapted to receive a second fluid. The protective cover **130** is also adapted to cooperate with the striker **140** and can be perforated by said striker **140** once this is displaced from a first starting position to a second opening position. FIG. 15 depicts, in a more detailed manner, the cooperation between the striker **140** and the protective cover **130**.

(46) FIG. 15 shows the fixing and use of the capsule **101** present inside a container **102**. A certain force is exerted onto the capsule **101** to fix said capsule **101** in the head **103** of the container **102**. When the capsule **101** is fixed, its edge **111** is positioned on the upper end of the head **103** of said container **102**.

(47) FIG. 15 diagrammatically depicts the at least partially deformed position of the fixing structures **121**, said position allowing said fixing structures **121** to leaktightly seal the space

between the exterior of the capsule **101** and the interior of the head of the container **102**.

(48) The container **102** is provided, within it, with a space **105** intended for receiving a first fluid. The capsule **101** is also provided with an inner space **106** adapted to receive a second fluid. For the obvious sake of clarity, the first fluid and the second fluid are not depicted in FIG. **15**. The space **106**, present inside the capsule **101**, is delimited by an assembly made up of the wall made of flexible material **150**, the upper protective cover **141**, the longitudinal wall **120** of the capsule **101** and the protective cover **130**. As shown in FIG. **15**, the striker **140** is provided, at its lower end, with a sharp edge.

(49) The protective cover **130** preferably has the same capacity to protect the second fluid as the protective cover **141** described above.

(50) The functionalities of the capsule **101**, as depicted in FIGS. **13**, **14** and **15**, are as follows.

(51) The container **102** provided with a capsule **101** can, before its first use, be sealed with the aid of a lid (not shown in FIG. **15**). A lid of this type can, for example, be fixed onto a thread **107** present on the exterior of the head **103** of the container **102**. During first use, the user removes the lid in order to access the upper part of the capsule **101**. As depicted with the aid of the arrow **160**, the user can exert a certain pressure onto the upper part of the striker **140** using his/her finger. This indicates that the user exerts a push onto the striker **140** at the protective cover **141**. By virtue of the presence of a wall made of flexible material **150**, the striker **140** can be displaced from the position as shown in FIG. **15** in the direction of the protective cover **130**. The user displaces the striker **140** until the protective cover **130** is pierced by the sharp edge of said striker **140**. When said protective cover **130** is pierced, the second fluid present within the capsule **101** is released and can flow within the space **105** and mix with the first fluid which is already present within said space **105**. The release of the second fluid towards the space **105** is depicted diagrammatically with the aid of the arrows **170**, which are visible in FIG. **15**.

(52) Once the first fluid and the second fluid have been mixed, as described in the paragraph above, the container **102** can be shaken in order to achieve an optimal mixing of the two fluids. During this mixing action, and by virtue of the presence of the protective cover **141** sealing the upper part of the striker **140**, there is no risk of leakage of said mixture in the container **102**.

(53) In order to release the mixture of the two fluids present in the container **102**, a following step consists of removing the protective cover **141**. This is depicted in FIG. **16**. When the protective cover **141** is removed, the upper end of the striker **140** is accessible to the user so that he/she can access the interior of the container **102**. Since the striker **140** substantially has a cylindrical form, this becomes, in a following step, a guide for an extraction device such as a pipette. The use of such a pipette is depicted in FIG. **17**.

(54) According to an alternative embodiment (not shown in the figures), the protective cover **141** can be adapted to be perforated with the aid of an extraction device.

(55) According to the example in FIG. **17**, a lid **108** is positioned on the outer end of the head **103** of the container **102**. The lid **108** seals the upper end of said container **102** and serves to guide a pipette **109** from the exterior to the interior of the container **102**. The pipette **109** is guided, with the aid of the lid **108** and the striker **140**, from the exterior of the container **102** to the interior in order to come into contact with the mixture of the fluids present inside the space **105** of the container **102**. The pipette **109** is provided with a head **191** which, according to one embodiment, can be made of a flexible material. Said flexible material allows the user to deform said head **191** in order to aspirate and extract a certain quantity of the fluid present inside said pipette **109**.

(56) The pipette **109** can be provided with a marking **192** which allows a specific sampling in line with one of the gradations of said marking **192**. When a specific quantity of fluid is extracted from the container **102** with the aid of the pipette **109**, said pipette can be repositioned in the lid **108**, thus sealing the upper end of the assembly of the container **102** and the lid **108** and making it possible to avoid the contents of said container **102** coming into contact with the ambient air.

(57) According to an alternative embodiment, not depicted in FIG. **17**, the upper part of the lid **108**

and the lower part of the head **191** of the pipette **109** can be provided with a “bayonet”-type system which allows said pipette **109** to be fixed in a defined position relative to the lid **108**.

## Claims

1. An assembly comprising a container, adapted to contain a first fluid, with an opening which allows the container to be filled and emptied, a lid adapted to seal the opening and a capsule to seal an opening in a container, wherein the container is adapted to contain a first fluid, the capsule being adapted to contain a second fluid that is different from the first fluid, the capsule comprising a striker which is movable from a first initial position to a second opening position and which, in the second opening position, allows the second fluid to be released into the container, thus mixing the first fluid and the second fluid, the striker being in a form of an open cylinder adapted to allow a fluid sampling device to pass through the capsule and to enter the container, wherein the capsule, in a use position of the capsule, at a side exposed towards an interior of the container, is defined by a first protective cover, a lower part of said striker being provided with a sharp end adapted to perforate the protective cover, wherein the container comprises fixing structures which allow the lid to be fixed onto the container, wherein an interior of the lid comprises at least one first push-piece adapted to push the capsule towards an interior of the opening in the container, thus fixing the capsule in the opening and wherein the interior of the lid comprises at least one second push-piece adapted to cooperate with a transverse wall of the striker, thus allowing the striker to be moved forward from its first initial position to its second opening position.
  2. The capsule according to claim 1, wherein the capsule comprises fixing structures for fixing the capsule in the opening in the container.
  3. The capsule according to claim 1, wherein the first protective cover comprises a material which is impermeable to oxygen.
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